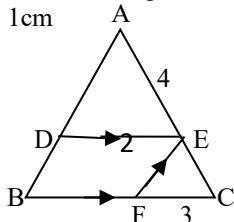
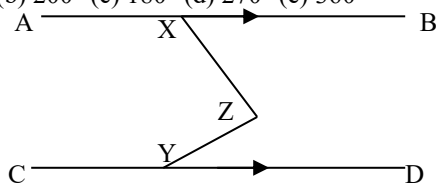


MATHEMATICS U.M.E 1978 QUESTIONS

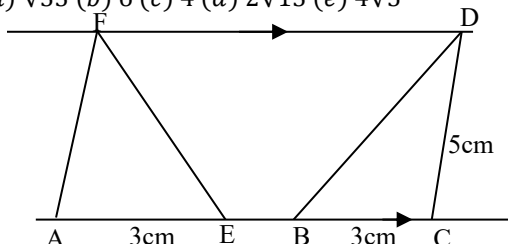
- A rectangular picture 6cm by 8cm is enclosed by a frame $\frac{1}{2}$ cm wide. Calculate the area of the frame. (a) 15sq.cm (b) 20sq.cm (c) 13sq.cm (d) 16sq.cm (e) 17sq.cm
- The sum of $3\frac{7}{8}$ and $1\frac{1}{3}$ is less than the difference between $3\frac{3}{8}$ and $1\frac{2}{3}$ by (a) $3\frac{2}{3}$ (b) $5\frac{1}{4}$ (c) $1\frac{1}{2}$ (d) 0 (e) $8\frac{1}{8}$
- Multiply $(x + 3y + 5)$ by $(2x^2 + 5y + 2)$ (a) $2x^3 + 3yx^2 + 10xy + 15y^2 + 13y + 10x^2 + 2x + 10$ (b) $2x^3 + 6yx^2 + 5xy + 15y^2 + 31y + 10x^2 + 2x + 10$ (c) $2x^3 + 3yx^2 + 5xy + 10y^2 + 13y + 5x^2 + 2x + 10$ (d) $2x^3 + 6yx^2 + 5xy + 15y^2 + 13y + 10x^2 + 2x + 10$ (e) $2x^3 + 2yx^2 + 10xy + 10y^2 + 31y + 5x^2 + 2x + 10$
- In the figure below $DE \parallel BC$, $DB \parallel FE$: $DE = 2$ cm, $FC = 3$ cm, $AE = 4$ cm. Determine the length of EC . (a) 4cm (b) 6cm (c) 3cm (d) 2cm (e) 1cm



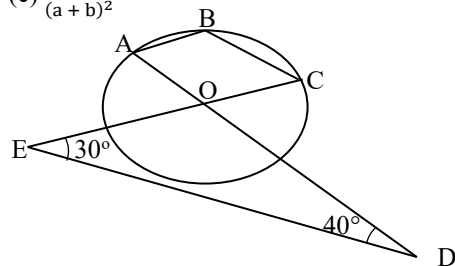
- In the figure below AB is parallel to CD , then $x + y + z =$ (a) 195° (b) 200° (c) 180° (d) 270° (e) 360°



- Arrange $\frac{3}{5}$, $\frac{9}{16}$, $\frac{34}{59}$ and $\frac{71}{97}$ in ascending order of magnitude (a) $\frac{3}{5}$, $\frac{9}{16}$, $\frac{34}{59}$, $\frac{71}{97}$ (b) $\frac{9}{16}$, $\frac{3}{5}$, $\frac{71}{97}$, $\frac{34}{59}$ (c) $\frac{9}{16}$, $\frac{34}{59}$, $\frac{3}{5}$, $\frac{71}{97}$ (d) $\frac{34}{59}$, $\frac{71}{97}$, $\frac{3}{5}$, $\frac{9}{16}$ (e) $\frac{3}{5}$, $\frac{9}{16}$, $\frac{71}{94}$, $\frac{34}{59}$
- The sum of the progression is $1 + x + x^2 + x^3 + \dots =$ (a) $\frac{1}{(1-x)}$ (b) $\frac{1}{1+x}$ (c) $\frac{1}{x-1}$ (d) $\frac{1}{x}$ (e) none of these
- The number of telephone calls N between two cities A and B varies directly as the population P_A , P_B in A , and B respectively and inversely as the square of the distance D between A and B . Which of the following equations represents this relation? (a) $N = \frac{kP_A}{D^2} + \frac{CP_B}{D^2}$ (b) $N = \frac{kP_AP_B}{D^2}$ (c) $N = kDP_AP_B$ (d) $N = kDP_A + CDP_B$ (e) $N = kD^2P_AP_B$
- Find the square root of $170 - 20\sqrt{30}$ (a) $2\sqrt{10} - 5\sqrt{3}$ (b) $3\sqrt{5} - 8\sqrt{6}$ (c) $2\sqrt{5} - 5\sqrt{6}$ (d) $5\sqrt{5} - 2\sqrt{6}$ (e) $5\sqrt{10} - 2\sqrt{3}$
- If $x^2 + 4 = 0$, then $x =$ (a) 4 (b) -4 (c) 2 (d) -2 (e) none of these
- In the figure below $FD \parallel AC$. The area of $AEF = 6$ sq. cm, $AE = 3$ cm, $BE = 3$ cm, $CD = 5$ cm, $\angle BCD$ is an obtuse angel. Find the length of BD . (a) $\sqrt{33}$ (b) 6 (c) 4 (d) $2\sqrt{13}$ (e) $4\sqrt{5}$



- What is the number whose logarithm to base 10 is 2.3482? (a) 223.6 (b) 0.2228 (c) 2.235 (d) 22.37 (e) 0.02229
- Five years ago, a father was 3 times as old as his son now, their combined ages amount to 110 years. Thus, the present age of the father is (a) 75years (b) 60 years (c) 98years (d) 81years (e) 105years
- If $y = 2x^2 + 9x - 35$. Find the range of values for which $y < 0$. (a) $-7 < x < \frac{5}{2}$ (b) $-7 \leq x < 5$ (c) $-5 \leq x < 7$ (d) $-\frac{7}{2} < x \leq 5$ (e) none of the above
- Father reduced the quantity of food bought for the family by 10% when he found that the cost of living had increased by 15%. Thus, the fractional increase in the family food bill is now (a) $\frac{1}{12}$ (b) $\frac{6}{35}$ (c) $\frac{19}{300}$ (d) $\frac{7}{200}$ (e) $\frac{5}{100}$
- Given that $a^*b = ab + a + b$ and that $a^0b = a + b + 1$. Find an expression (not involving $*$ or 0) for $(a^*b)^0(a^*c)$ if a, b, c are real numbers and the operations on the right are ordinary addition and multiplication of numbers. (a) $ac + ab + bc + b + c + 1$ (b) $ac + ab + a + c + 2$ (c) $ab + ac + a + b + 1$ (d) $ac + bc + ab + b + c + 2$ (e) $ab + ac + 2a + b + c + 1$
- If a circular paper disc is trimmed in such a way that its circumference is reduced in the ratio 2: 5, in what ratio is the surface area reduced? (a) 8: 125 (b) 2: 5 (c) 8: 25 (d) 4: 25 (e) 4: 10
- If the four interior angles of a quadrilateral are $(P + 10)^\circ$, $(P - 30)^\circ$, $(2P - 45)^\circ$, and $(P + 15)^\circ$, then P is (a) 125° (b) 82° (c) 135° (d) 105° (e) 60°
- Simplify $\frac{a-b}{a+b} - \frac{a+b}{a-b}$ (a) $\frac{4ab}{a^2-b^2}$ (b) $\frac{2ab}{a^2-b^2}$ (c) $\frac{-4ab}{a^2-b^2}$ (d) $\frac{4ab}{(a-b)^2}$ (e) $\frac{2ab}{(a+b)^2}$



- In the figure above, O is the centre of the circle ABC , $\angle CED = 30^\circ$, $\angle EDA = 40^\circ$. What is the size of ABC ? (a) 70° (b) 110° (c) 130° (d) 125° (e) 145°
- A force of 5 units on a particle in the direction to the east and another force of 4 units acts on the particle in the direction north- east. The resultant of the two force is (a) 3 units (b) $\sqrt{3}$ units (c) $\sqrt{41 + 20\sqrt{2}}$ units (d) $\sqrt{41 - 20\sqrt{2}}$ unit (e) $\sqrt{41\sqrt{2} + 20}$ units
 - The minimum point on the curve $y = x^2 - 6x + 5$ is at (a) (1,5) (b) (2,3) (c) (-3, -4) (d) (-3, 4) (e) (3, -4)
 - If $3x - \left(\frac{1}{4}\right)^{\frac{1}{2}} > \frac{1}{4} - x$, then the interval of values of x is (a) $x > \frac{1}{3}$ (b) $x < \frac{1}{3}$ (c) $x < \frac{1}{4}$ (d) $x < \frac{9}{16}$ (e) $x > \frac{9}{16}$
 - A canal has rectangular cross section of width 10cm and breadth 1m. If water of uniform density 1gm cm^{-3} flows through it as a constant speed of 1000mm per minute, the adjacent sea is... (a) 100 000 (b) 1000 000 (c) 120 000 (d) 30 000 (e) 350 000
 - A man runs a distance of 9km at a constant speed for the first 4km and then 2km/h faster for the rest of the distance. The

- whole run takes him one hour. His average speed for the first 4km is (a) 6km/h (b) 8km/h (c) 9km/h (d) 11km/h (e) 13km/h
26. A pyramid is constructed on a cuboid. The figure has (a) twelve faces (b) thirteen vertices (c) fourteen edges (d) fifteen edges (e) sixteen edges
27. In a geometric progression the first term is 153 and the sixth term is $17/27$. The sum of the first four terms is (a) $860/3$ (b) $680/3$ (c) $608/3$ (d) $806/3$ (e) $680/27$
28. An arithmetic progression has first term 11 and fourth term 32. The sum of first nine terms is (a) 351 (b) 531 (c) 135 (d) 315 (e) 153
29. A triangle has angles 30° , 15° , and 135° . The side opposite to the angle 30° is length 6cm. The side opposite to the angle 135° is equal to (a) 12cm (b) 6cm (c) $6\sqrt{2}$ cm (d) $12\sqrt{2}$ cm (e) $6\sqrt{3}$ cm
30. A regular hexagon is constructed inside a circle of diameter 12cm. The area of the hexagon is (a) $36\pi\text{cm}^2$ (b) $36/\pi\text{cm}^2$ (c) $54\sqrt{3}\text{cm}^2$ (d) $54/\sqrt{3}\text{cm}^2$ (e) $54\sqrt{3}x\text{cm}^2$
31. In a soccer competition in one season, a club had scored the following goals: 2, 0, 3, 3, 2, 1, 4, 0, 0, 5, 1, 0, 2, 2, 1, 3, 1, 4, 1, and 1. The mean, median and mode are respectively (a) 1.18 and 1.5 (b) 1.8, 1.5 and 1 (c) 1.8, 1 and 1.5 (d) 1.5, 1 and 1.8 (e) 1.5, 1.8 and 1
32. In $\sec^2 \theta + \tan^2 \theta = 3$, then the angle θ is equal to (a) 30° (b) 45° (c) 60° (d) 90° (e) 105°
33. A hollow right prism of equilateral triangle base of side 4cm is filled with water up to a certain height. If sphere of radius $1/2$ cm is immersed in the water, then the rise of water is (a) 1cm (b) $\sqrt{3}\pi/24$ cm (c) $\pi/24\sqrt{3}$ cm (d) $24\pi/\sqrt{3}$ cm (e) $24\sqrt{3}\pi$ cm
34. The set of values of x and y which satisfies the equations $x^2 - y - 1 = 0$ and $y - 2x + 2 = 0$ is (a) 1, 0 (b) 1, 1 (c) 2, 2 (d) 0, 2 (e) 1, 2
35. A solid sphere of radius 3cm, a solid right cone of radius 3cm and height 12cm and a solid right circular cylinder of radius 3cm and height 4cm. Which of the following statements is true? (a) The volume of the sphere is greater than the volume of the cone (b) the volume of the cone is less than the volume of the cylinder (c) the total surface area of the cone is greater than that of the sphere (d) the total surface area of the cylinder is less than that of the sphere (e) the total surface area of the cone is equal to that of the cylinder
36. The quantity $(x + y)$ is a factor of (a) $x^2 + y^2$ (b) $x^3 - y^3$ (c) $2x^2 - 3xy + y^2 - x + 1$ (d) $2x^3 + 2x^2y - xy + 3x - y^2 + 3y$ (e) $x^5 - y^5$
37. Two triangles have the same area if (a) two sides in one triangle are equal to two sides in the other (b) three sides in one triangle are equal to three sides in the other (c) two angles in one triangle are equal to two angles in the other (d) three angles in one triangle are equal to three angles in the other (e) one side and the opposite angle in one triangle are equal to one side and the opposite angle in the other
38. If $(25)^{x-1} = 64(5/2)^6$ then x has the value (a) 7 (b) 4 (c) 32 (d) 64 (e) 5
39. Assuming $\log_e 4.4 = 1.4816$ and $\log_e 7.7 = 2.0142$ then the value of $\log_e \frac{1}{4}$ is (a) 0.5326 (b) 3.4958 (c) 0.4816 (d) 0.0142 (e) 1.3594
40. The locus of all points having a distance of l units from each of the two fixed points a and b is: (a) a line parallel to the line ab (b) a line perpendicular to the line ab through the mid-point of ab (c) a circle through a and b with centre at the mid-point of ab (d) a circle with centre at a and passes through b (e) a circle in a plane perpendicular to ab and centre at the mid-point of the line ab.
41. Without using tables simplify $\frac{\ln\sqrt{216} - \ln\sqrt{125} - \ln\sqrt{8}}{2(\ln 5 - \ln 3)}$ (a) -3 (b) 3 (c) $3/4$ (d) $-3/4$ (e) $\ln 6 - 2 \ln 5$
42. Simplify $\frac{(a^{-1}/a)(a^{4/3+1/a^{2/7}})}{(a^2 - 1/a^2)}$ (a) $a^{2/3}$ (b) $a^{-1/3}$ (c) $a^2 + 1$ (d) a (e) $a^{1/3}$
43. If $x^4 - kx^3 + 10x^2 + lx - 3$ is divisible by $(x - 1)$, and if when it is divided by $(x + 2)$ the remainder is 27, find the constants k and l . (a) $k = -7, l = -15$ (b) $k = -15, l = -7$ (c) $k = -21/5, l = -61/5$ (d) $k = -5/3, l = 19/3$ (e) $k = 5/3, l = -19/3$
44. Evaluate without using tables $\sin(-1290^\circ)$ (a) $\sqrt{3}/2$ (b) $-\sqrt{3}/2$ (c) $\sqrt{2}/2$ (d) 1 (e) $1/2$
45. The angle of elevation of the top of a vertical tower from a point A on the ground is 60° . From a point B, 2 units of distance further away from the foot of the tower, the angle of elevation of the top of the tower is 45° . Find the distance of A from the foot of the tower. (a) $3 + \sqrt{3}$ (b) $5 + \sqrt{3}$ (c) $3 - \sqrt{3}$ (d) $1 + \sqrt{3}$ (e) $\sqrt{3} - 1$